Verdandi

norns on the Mac: a community port, as a standalone app and an AUv3 plugin, with the script catalog. not affiliated with monome. free software under the GPL-3.0.

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1. what this is

norns is a sound computer by monome: a small box running Lua scripts, a SuperCollider audio engine and the softcut sample looper, with a 128×64 screen, three keys and three encoders. Verdandi runs the same software natively on macOS. the window shows the device: the screen behind its bezel, the keys and encoders where they sit on a norns, played with the mouse or the computer keyboard. the community scripts run unchanged.

the name: norns is monome's, and this port is not theirs, so it has a name of its own. Verdandi is one of the three norns of the old stories, the one of the present, of what is happening now.



everything that makes a norns a norns is here: matron (scripts, screen, devices), crone and softcut (audio routing, loopers, tape), SuperCollider with the sc3-plugins, and maiden (the web editor with the script catalog). in addition Verdandi installs an AUv3 plugin that puts norns into a DAW track.

new to norns? norns in five minutes has the basics; monome's norns documentation has the rest, and it applies to Verdandi too.

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2. installing

requirements: a Mac with Apple silicon (M1 or later), macOS 15 Sequoia or newer.

1. open `Verdandi.dmg` and drag **Verdandi** onto **Applications**.
2. the app is not notarized by Apple (it is ad-hoc signed, without a developer identity), so macOS blocks the first start. open Verdandi once and close the message, then open *System Settings* → *Privacy & Security*, scroll down to Verdandi and click **Open Anyway**, and confirm. (macOS 15 no longer offers right-click → Open for this.) if macOS says the app is damaged, run `xattr -dr com.apple.quarantine /Applications/Verdandi.app` in Terminal.
3. when Verdandi asks for the microphone, allow it: that is the audio input for softcut, tape and engines. (you can change it later under *Privacy & Security* → *Microphone*.)

updates: when Verdandi starts (and once a day while it runs) it asks GitHub whether there is a newer release, and if so says it on the window: **Download** opens the release page, **Later** asks again at the next start, **Skip This Version** waits for the one after. it sends nothing but the app's version. to update, drag the new Verdandi onto Applications and replace the old one; dust in `~/norns` stays as it is. *Verdandi* → *Check for Updates Automatically* turns the check off.

3. first start

Verdandi starts the audio server, the SuperCollider engine and maiden, then opens its window with the norns device. the first start takes a little longer: macOS checks the bundled libraries once.

your norns lives in the folder `~/norns` in your home folder, laid out like `/home/we` on the device:

path	contents
`~/norns/dust/code`	installed scripts
`~/norns/dust/audio`	samples, tape recordings (`audio/tape`)
`~/norns/dust/data`	script data, PSETs, system settings
`~/norns/norns`	a link into the app (the norns core); do not edit
`~/norns/*.log`	logs, see files and logs

to stop Verdandi, close its window, quit it from the Dock, or choose SLEEP in the norns menu. only one Verdandi runs at a time.

4. norns in five minutes

Verdandi plays like a norns. what monome's norns documentation says holds here too, apart from the few things in differences from the device. the short version:

- **a script plays, the menu waits.** norns runs one script at a time, and the screen, keys and encoders belong to it. a short press on K1 (`1`) switches between the script and the menu.
- **in the menu** E1 goes through the pages MIX, TAPE, HOME and PARAMS (the buttons above the device do the same), E2 scrolls, K3 enters or confirms, K2 goes back.
- **HOME:** SELECT picks a script (K3 loads it), SYSTEM has devices, audio, mods, reset and restart, SLEEP quits Verdandi.
- **PARAMS:** everything the script lets you set, then norns' own settings (levels, reverb, compressor, clock). PSET saves and loads them as presets.
- **MIX:** the levels of out, in, monitor, engine, softcut and tape.
- **TAPE:** records what norns plays into `dust/audio/tape` , and plays files back.
- **new scripts** come from the catalog in maiden; then SELECT them. how a script is played is at the top of its file, in maiden's catalog entry, or on its page on lines.
- **K1 + turn:** many scripts do something else while K1 is held: hold `1` and turn.

more, from monome: playing norns (the menus in depth), maiden, clocks and controllers,

writing scripts, the API reference and help. questions about Verdandi itself go to its GitHub issues, not to monome.

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5. controls

keyboard

while the Verdandi window is in front, the keyboard is the norns hardware:

keys	norns
`1` `2` `3`	K1 K2 K3 (held while you hold the key, so K1 + turn works)
`↩` (Return) or `Space`	K3: confirm
`Esc` or `⌫` (Backspace)	K2: back
`Q` / `W`	E1 left / right
`A` / `S`	E2 left / right (in scripts also `↑` / `↓`)
`Z` / `X`	E3 left / right (in scripts also `←` / `→`)
with `⇧`	5 steps at once; holding a key keeps turning
`F1`–`F4`, `F5`	menu pages MIX, TAPE, HOME, PARAMS; F5 toggles the menu
`F6` `F7` `F8`	open or close the virtual grid, keyboard and arc

this works wherever nothing else wants the keys. when a text entry is open (naming a file, searching), you type. scripts that use the keyboard themselves (live coding, keyboard instruments, about 14 of the 350 catalog scripts) get the keys; there, hold `⌥` (Option) with the keys above, or press `⌥` `Esc` to force hardware keys on or off (the window title then reads *Verdandi · keys*).

mouse

the keys and knobs in the window are the controls:

- **keys:** click K1, K2 or K3. the key stays down while you hold the mouse button.
- **knobs:** drag a knob up or right to turn it clockwise, down or left for counter-clockwise. or scroll over it. scrolling anywhere else turns the nearest knob. hold `⇧` while dragging for fine steps, `⌘` for fast ones.
- for K1 + turn, hold `1` on the keyboard and turn a knob with the mouse.
- **the MIX page:** drag a level bar on the screen up or down (out, in, mon, eng, cut, tp), or scroll over it; a double click sets it back to 0 dB.

- **OUT and MUTE** (right of E3, not on the real norns): the fader is norns' output level, the same as *out* on the MIX page: drag it or scroll over it, double click it for 0 dB. MUTE silences the output and brings it back at the level it had; turning the level up also ends the mute. norns starts with the level from before the mute.

the keys and knobs in the window always show what norns receives, also from the keyboard.

the buttons above the device

the real norns does not have these; they save trips through the menu:

button	does
MIX TAPE HOME PARAMS	opens that menu page (its light shows the page you are on). click the lit one to go back to the script.
MIC	chooses what norns' input hears; each click goes on to the next: **MIC** (the Mac's microphone or your interface's inputs), **SYS** (the sound of every other app: a video in the browser, a synth, Music) and **OFF**. **APP** means one app chosen in Devices → Audio Input, see other apps as input. its light is on while something reaches norns. switch the microphone off when speakers are loud and close to it, so they cannot feed back. norns starts with what it had.
GRID ARC KEYS	open or close the virtual grid, arc and keyboard (also `F6`, `F8`, `F7`), see devices. the light is on while one is open. norns and the open devices are laid out together so all fit the screen; when the last one closes, norns goes back to where it was.
MAIDEN	opens maiden in your browser.
MANUAL	opens this manual.

the menu bar

menu	does
Verdandi → Check for Updates…	asks GitHub for a newer Verdandi now (see updates); *Check for Updates Automatically* below it turns the check at start and once a day on or off.
File → Import Audio or Script… ⌘O	adds audio and scripts to norns, like dropping them on the window (see files); tick *Copy audio into norns* for a copy instead of a link.
File → Show dust Folder ⇧⌘D	opens norns' dust folder (scripts, audio, data) in the Finder.
File → Show Tape Recordings ⇧⌘T	opens the folder TAPE records into.
File → Save Screenshot ⇧⌘S	saves the screen as a sharp 512×256 picture on your desktop, for sharing a script on lines or norns.community.
File → Record GIF ⇧⌘G	records the screen as an animated GIF (up to a minute, a red dot shows it is recording); the same item stops it and saves it on your desktop.
Devices	opens or closes the grid, keyboard and arc.
Devices → Audio Input	what norns' input hears, as with the MIC button, or one app of those running; see other apps as input.
View → Device Skin / Dark Flat Skin	how the windows look: as the aluminium device, or dark and flat (easier on the eyes in a dark room). the grid, arc and keyboard follow; the choice is kept for the next start.

window

the window can be resized and keeps the shape of the device; the screen is always drawn with sharp pixels. **View → Dark Flat Skin** draws it dark and flat instead of aluminium.

6. audio

norns starts with the Mac's **default input and output** (*System Settings → Sound*) at 48 kHz with a 256-frame buffer (about 5 ms). to use an audio interface, another sample rate or a smaller buffer, go to **SYSTEM → AUDIO**:

- **E2** picks a line, **E3** changes it: INPUT, OUTPUT (an interface or *default*), RATE (44.1 to 96 kHz), BUFFER (64 to 1024 frames, with the latency next to it).
- go to APPLY and press **K3**: norns restarts its audio with the new setting, which is kept for the next start.
- if the setting does not work (say the interface cannot do that rate), norns goes back to the defaults and the AUDIO page says so. an interface that is unplugged is replaced by the Mac's default device.

a mono microphone (like the built-in one) feeds both norns inputs. levels, monitoring, reverb and compressor are in the norns MIX menu, as on the device.

other apps as input

to loop, sample or process what other apps play (a video in the browser, a synth, a drum machine, Music) without a DAW or a loopback driver, click MIC until it says **SYS**: norns' input hears the sound of every app but norns itself. for one app only, choose it in **Devices → Audio Input** (apps that are playing right now are at the top); the button says **APP**.

- the sound goes into norns' inputs instead of the microphone or interface. scripts record and process it like anything plugged into a norns: record a buffer in a looper while a video plays, and it is in the buffer.
- the apps keep playing as before: you hear them directly, and norns gets a copy. norns' input monitor of them goes to TAPE only, so nothing is heard twice: TAPE records the video together with what norns plays, softcut and engines get it as their input.
- the app does not have to play, or even run, yet: norns picks it up when it starts.
- the first time, macOS asks whether Verdandi may record audio from other apps; allow it. if the app stays silent in norns, check *System Settings → Privacy & Security → Screen & System Audio Recording*.
- while a DAW plays through the AUv3 plugin, norns' input is the DAW track: SYS and APP pause meanwhile, and every app (the DAW too) is heard directly.

one app at a time. SYS and APP are kept for the next start; a note on the device says so.

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7. maiden and the script catalog

maiden is norns' web editor. with Verdandi running, open `http://127.0.0.1:5000/maiden` in your browser.

- **project manager** (the box icon): browse the community catalog, install and update scripts. installed scripts appear in the norns SELECT menu.
- **editor**: edit everything in `dust`.
- **REPL**: the matron (Lua) and SuperCollider consoles; errors from scripts show up here.

unlike on the device, maiden and the consoles only answer on this Mac (127.0.0.1), not on your network: the consoles can run any code.

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8. scripts

scripts run unchanged. the port takes care of what differs from the device:

- scripts see the usual `/home/we/dust/…` paths; norns maps them to `~/norns`.
- the sc3-plugins (Decimator, MoogLadder, Greyhole, …) are built in.
- `require "cjson"` uses the built-in JSON library.
- shell commands find the same GNU tools as on the device.

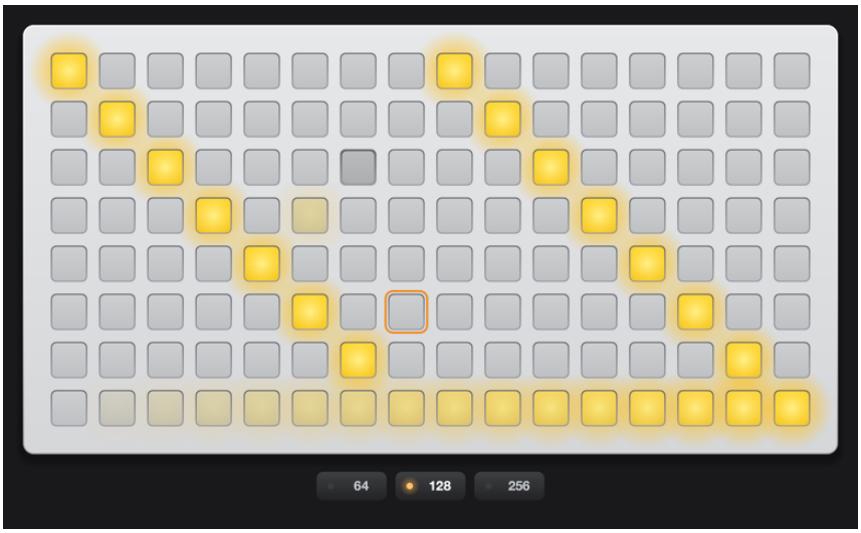
scripts that bring compiled Linux programs or Linux-only UGens, or that install software with `apt`, can fail in those parts; everything else in them still works. an engine that fails to load blocks later engine loads until you restart norns (as on the device).

new engines are compiled when norns starts: after installing a script with its own engine, restart norns (or use SYSTEM → RESTART in the menu).

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9. MIDI, grid, arc, crow, gamepads

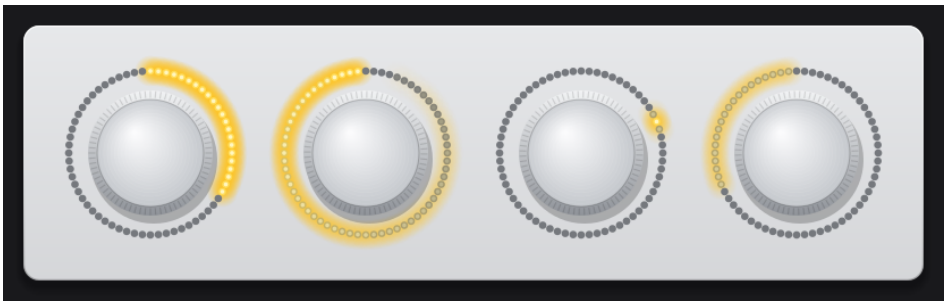
- **MIDI**: class-compliant USB MIDI devices and network/IAC ports appear in SYSTEM → DEVICES → MIDI. norns also creates a virtual port **norns** that other apps (a DAW) can send to and receive from.
- **grid and arc**: plug them in; no serialosc needed.
- **no grid?** the GRID button (or `F6`) opens a virtual grid in its own window, as a 64 (8×8), 128 (16×8) or 256 (16×16): pick the size with the switch under it (norns keeps the choice; a script that is running may need to be loaded again to use a new size). norns and the scripts see it as a real grid (in SYSTEM → DEVICES it is *virtual grid*). click a key to press it, drag across keys to play them in turn, right-click (or ctrl-click) to hold a key down, for chords; right-click again to let it go. closing the window unplugs it.



- **no MIDI keyboard?** the KEYS button (or `F7`) opens a two octave keyboard. norns and the scripts see it as the MIDI device *Verdandi keys* (other apps, a DAW, see it too); it takes the first free port in SYSTEM → DEVICES → MIDI, so a synth script that listens to port 1 plays it right away when nothing else is plugged in. click a key to play it: near the front is loud, further back is soft. drag across keys to play them in turn. right-click (or switch on LATCH) to hold notes, for drones and chords. with the keyboard window in front, the computer keyboard plays it like a piano: `A` is C, the row `A`–`'` the white keys, `W` `E` `T` `Y` `U` `O` `P` the black ones (by position, on any keyboard layout; the keys show their letters). `Z` / `X` move the octave, `C` / `V` the velocity.



- **no arc?** the ARC button (or `F8`) opens a virtual arc with four rings; scripts see it as an arc (in SYSTEM → DEVICES it is *monome arc virtual*). drag round a knob to turn it (clockwise is up; a whole turn of the mouse round it is a quarter turn of the knob, so it can be set finely), or scroll over it; hold `⇧` for fine turns, `⌘` for fast ones. right-click a knob to push it.



- **crow**: plug it in; it appears like on the device.
- **gamepads / joysticks**: USB and Bluetooth controllers work with scripts that support them.
- **keyboard**: the Mac keyboard is the norns keyboard (see controls).

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10. the AUv3 plugin

Verdandi installs an Audio Unit called **Verdandi** (a music effect: audio in, audio out, MIDI in). it puts norns into a DAW track: the track's audio goes into norns, and norns' output comes back into the track.

1. start **Verdandi** (the plugin talks to it; norns keeps running in its own window).
2. in your DAW, insert **Verdandi** on a track — in Logic as an Audio FX or MIDI-controlled effect, in Ableton Live / Bitwig / Reaper from the Audio Unit plugins.
3. press play. the plugin window shows *connected*. while the DAW plays through the plugin, norns hears the track instead of your audio interface and plays into the track instead of your speakers. when the DAW stops, norns goes back to the interface.
4. MIDI sent to the plugin track reaches norns as the MIDI device **Verdandi AUv3** (choose it in a script's MIDI settings or in SYSTEM → DEVICES).

latency: about 8 ms, reported to the DAW so it compensates. norns follows the DAW's clock and sample rate (it resamples internally). use one Verdandi plugin at a time. if Verdandi is not running, the plugin passes audio through unchanged.

if the plugin does not show up: start Verdandi once from Applications, then rescan plugins in the DAW (Logic: Plug-in Manager → Reset & Rescan).

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11. files and logs

getting samples and scripts in: drop files or folders on any norns window (or choose File → Import). a note on the device says what happened.

a sample for a script: when a script asks for a file (say *buffer 1 sample* in PARAMS), the Finder's open panel comes up next to norns' own file browser: choose the sample anywhere on your Mac, and the script gets it (linked into `dust/audio/mac`, see below). or drop it on the norns window while the browser is up. *Cancel* leaves norns' browser to

choose from dust as on the device.

- **audio** (wav, aiff, flac, ogg, mp3, and folders with audio in them) is *linked*: the files stay where they are, and `dust/audio/mac` gets a link to them, where sampler scripts find them like any other folder. nothing is copied, so there is nothing to clean up: deleting a link in `dust/audio/mac` never deletes your samples, and links whose files were moved or deleted are removed the next time you add something. if you move the originals or unplug their disk, norns cannot play them until they are back.
- hold ⌥ (Option) while dropping to *copy* the audio into `dust/audio` instead: for samples a saved preset should always find.
- **scripts** (a folder with a `.lua` file in it, or a single `.lua` file) are always copied to `dust/code`, because they belong to norns: maiden updates them, and norns starts the last one again on the next launch.
- **from other apps** (Voice Memos, Music, DAWs, sample browsers): many hand over a file only when it is dropped, or the sound itself. that is copied into `dust/audio`, as there is no original to link to; so is a file from a temporary folder, which the app would delete. if an app drags nothing norns can use, the note on the device says so: export the sound as a file first, or play it in with SYS or Audio Input.
- nothing is overwritten: a name that is already there gets a number.

getting things out: TAPE recordings are in `dust/audio/tape` (File → Show Tape Recordings); screenshots and GIFs of the screen go to your desktop.

file	what
`~/norns/norns.log`	matron and crone (Lua errors, devices)
`~/norns/sclang.log`	SuperCollider (engines, UGens)
`~/norns/jackd.log`	audio server
`~/norns/maiden.log`	maiden
`~/norns/auv3_bridge.log`	the AUv3 connection

to reset norns' settings, choose SYSTEM → RESET (or quit Verdandi and delete `~/norns/dust/data/system.*`). your scripts and audio are untouched.

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12. troubleshooting

problem	what to do
a script says "version … required"	it needs a newer norns than this Verdandi contains: a new version of Verdandi brings it. (Verdandi before 0.1.1 always said 000000 here.)
"Verdandi cannot be opened"	*System Settings* → *Privacy & Security* → *Open Anyway* (see installing).
SUPERCOLLIDER FAIL on the first start	macOS was still checking the bundled libraries; quit and start Verdandi again.
no sound	check the default output in System Settings → Sound, the levels in the MIX menu, and `jackd.log`.
no input	allow the microphone for Verdandi (Privacy & Security → Microphone); check the input level in MIX.
a second Verdandi does not start	only one at a time; close the other window first.
a script misbehaves	open maiden's REPL or `norns.log` for the error.
maiden does not load	use `http://127.0.0.1:5000/maiden` (not `localhost`), with Verdandi running.

13. differences from the device

- SYSTEM shows DEVICES, AUDIO, MODS, RESET and RESTART. Wi-Fi and the password are the Mac's job, scripts are updated in maiden, and Verdandi itself with a new version of the app. the norns release it contains (scripts that ask for a newer norns check it) is written to `~/norns/version.txt` at every start; this one is 260819.
- the encoders are endless: E1 goes round the menu pages (after PARAMS comes MIX again). inside a page and in scripts, values stop where the page or script lets them stop.
- SYSTEM → RESTART restarts norns and SuperCollider; SLEEP quits Verdandi (it does not put the Mac to sleep).
- no battery or screen settings: there is no battery, and the screen is a window.
- audio device and sample rate come from macOS (see audio).
- maiden and the consoles are local-only (127.0.0.1); OSC (port 10111) is reachable on your network, so phone controllers like TouchOSC work as on the device.

14. uninstalling

delete `/Applications/Verdandi.app`. your norns folder `~/norns` (scripts, audio, data) stays; delete it too if you no longer need it.

15. source and licenses

Verdandi is free software. the app contains, each under its own license: norns (GPL-3.0, monome and contributors), maiden (GPL-3.0), SuperCollider (GPL-3.0), sc3-plugins (GPL-2.0 or later), JACK (GPL-2.0 / LGPL-2.1), Lua (MIT), lua-cjson (MIT), liblo (LGPL-2.1), libmonome (ISC), cairo (LGPL-2.1 / MPL-1.1), SDL (zlib), libsndfile (LGPL-2.1), libsamplerate (BSD-2), nng (MIT), Boost (BSL-1.0), yaml-cpp (MIT), GNU coreutils, findutils, sed and grep (GPL-3.0), and their dependencies. the folder **Licenses** (in the dmg next to the app, and in `Verdandi.app/Contents/Resources/Licenses`) has the license texts of every part and `PACKAGES.txt`, the list of all of them with version, license and where the exact source is. the complete source of the port, with build instructions (`readme-macos.md`, `macos/setup-deps.sh`, `macos/make-app.sh`), is published with each release; GPL-3.0 grants you the right to get, study, change and share it.

Verdandi runs norns, made by monome and its community. norns and monome are names of monome; Verdandi is an independent community project, not endorsed by or affiliated with monome.